

GURUKULA KANGRI (DEEMED TO BE UNIVERSITY), HARIDWAR
END SEMESTER EXAMINATION-2021

CLASS – B.Tech

SEMESTER – III

SUBJECT NAME – Engineering Mathematics-III

SUBJECT CODE- BEM-C302

TIME: 3 HOUR

MAX. MARKS: 70

MIN. PASS: 40%

NOTE: QUESTION PAPER IS DIVIDED INTO TWO SECTIONS A AND B. ATTEMPT BOTH SECTIONS AS PER GIVEN INSTRUCTIONS.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

INSTRUCTIONS: ANSWER ANY FIVE QUESTIONS IN ABOUT 150 WORDS EACH. EACH QUESTION CARRIES SIX MARKS. (5 X 6 = 30 MARKS)

Q.1. Find inverse Laplace transform of $\tan^{-1} \frac{2}{s^2}$.

Q.2. Find the value of $\int_0^\infty \frac{\cos 6t - \cos 4t}{t} dt$

Q.3. Find $f(x)$ whose Fourier cosine transform is $\frac{\sin a\lambda}{\lambda}$.

Q.4. Evaluate $\int_0^\infty \frac{x^2}{(x^2+a^2)(x^2+b^2)} dx$ and hence find $\int_0^\infty \left(\frac{x}{1+x^2}\right)^2 dx$.

Q.5. Find the inverse Z transform of $\frac{4z^2-2z}{z^3-5z^2+8z-4}$.

Q.6. Determine u_2 where $U(z) = \frac{2z^2+3z+4}{(z-3)^3}$.

Q.7. Show that the function $w = \frac{x}{x^2+y^2} - \frac{iy}{x^2+y^2}$ is an analytic function. Find $\frac{dw}{dz}$.

Q.8. Evaluate $\int_C z^2 dz$ where C is a

(i) Straight line segment from $z=0$ to $z=2+i$

(ii) Straight line from $(0, 0)$ to $(2, 0)$ and then from $(2, 0)$ to $(2, 1)$.

Q.9. Use Newton Raphson method to find the real root of the equation $3x = \cos x + 1$, correct to 5 decimal places.

Q.10. Use iteration method to find root of equation $x^3 + x^2 - 1 = 0$, correct to 4 decimal places.

SECTION-B (LONG ANSWER TYPE QUESTIONS)

INSTRUCTIONS: ANSWER ANY FOUR QUESTIONS IN DETAIL. EACH QUESTION CARRIES TEN MARKS. (4 X 10 = 40 MARKS)

Q.1. Use Laplace transform to solve $y' + y - 2 \int_0^t y dt = \frac{t^2}{2}$, $y(0) = 1, y'(0) = -2$.

Q.2. Find the Fourier cosine transform of $f(x) = \frac{1}{1+x^2}$. Hence find the Fourier sine transform of $\frac{x}{1+x^2}$.

Q.3. Solve the integral equation $\int_0^\infty f(x) \cos \lambda x dx = \begin{cases} 1-\lambda & 0 < \lambda < 1 \\ 0 & \lambda > 1 \end{cases}$

Hence show that $\int_0^\infty \frac{\sin^2 u}{u^2} du = \frac{\pi}{2}$.

Q.4. Use Z transform to solve $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$ given $y_0 = y_1 = 0$.

Q.5. Use Cauchy integral theorem to evaluate $\oint_C \frac{e^z}{z(1-z)^3} dz$ where C is

(a) $|z + i| = \frac{1}{2}$ (b) $|z - 1| = \frac{1}{2}$ (c) $|z| = 2$.

Q.6. Use Milne Thompson method to construct $f(z) = u + iv$

where $u = e^x(x \cos y - y \sin y) + 2 \sin x \cdot \sinh y + x^3 - 3xy^2 + y$.

Q.7. Use Cauchy Residue theorem to evaluate $\int_0^{2\pi} \frac{d\theta}{2 + \cot \theta}$.

Q.8. Use bisection method to approximate the real root of $e^{-x} - x = 0$ correct to 4 decimal places.

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